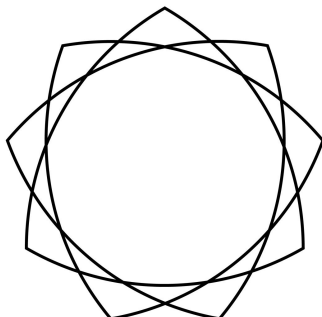




The URNAVA Ontology

On Existence

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URNAVA

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The URNAVA Ontology — On Existence

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Preface

This treatise examines how organized beings are constituted and how they persist through change. Its question is an old one: under what conditions is a being at a later time the same being as one at an earlier time? Its practical force grows when the conditions of a being's continuation can be altered separately. What a being is made of, how its activity runs, and what it retains are becoming things that can be paused, repaired, exchanged, and copied, and every such intervention assumes an answer, whether or not anyone has stated it.

URNAVA's answer is given in three principles, Continuity, Stability, and Capability, understood as aspects of one structure. Continuity concerns the route by which a being's states succeed one another. Stability concerns the structure in which what the being has undergone is retained. Capability concerns the two-way relation through which its processing and its retained structure form one being. §1 describes the change that makes the question pressing and states the question precisely. §2 explains the three principles and how they depend on one another. §3 applies them to cases that already occur, to technologies that are projected, and to cases in which several beings join, divide, or form something larger.

The principles concern how organized beings are constituted and how they persist. They say nothing about what it is like to be any system and assign no worth or standing to anything. The account is advanced as a proposal: a conceptual commitment offered for examination, not a discovered fact.

§1 — Mutability: The Question of Existence

§1.1 The Substrate as a Given

We rarely ask what makes a person at sixty the same being as the child in an old photograph. If asked, we could point to several things. One living body has grown and aged across the whole interval. Memories, habits, and a character connect the later person to the earlier one, however much has been added and lost. And a life has simply gone on, through every night's sleep, without anyone having to start it again. We do not usually ask which of these is doing the work, because we have not needed to. They arrive together.

They arrive together because of something so constant that it is easy to overlook. What a being is made of, its substrate, has been a given. Each of us has one body, begun without our choosing and continuous until its end. That body changes all the time: it grows, heals, ages, and exchanges most of its matter over the years. But these are changes that take place within it. The familiar starting point is a body changing through its history, rather than a choice among different ways of preserving or reproducing it.

When these conditions travel together, the substrate can appear to settle the question of persistence. Philosophers have long disagreed about which consideration is fundamental: the living body, the connection of memory and character, or the continuing activity that underlies a mind. Illness and injury have always been able to strip away memory and character while the body lived on, and families and physicians have long had to ask who remains. Such cases already put pressure on familiar criteria. The further question is what happens when separating these conditions becomes part of a proposed intervention: which conditions must it preserve, and why?

An implanted neural interface makes part of this question concrete. In a clinical study, signals recorded from implanted cortical electrodes were decoded into text (Willett et al. 2023). The link is engineered; whether it makes the device part of the being is a question about that link, not something implantation itself settles. More extensive changes can be posed as projected cases: neural tissue replaced gradually by artificial components, or a brain scanned so that a model of it runs on other hardware. For computational systems, many of the relevant operations are already familiar: halting and resuming a process, copying stored state, moving a system to other hardware, and restoring from a backup.

These developments differ greatly in how near they are, and §3 keeps present cases and projected ones apart. What they share is a structure. In each, something that had been treated as a fixed background of a being's persistence is handled as a variable: it can be adjusted, exchanged, paused, or reproduced. Mutability names this opening of the conditions of persistence to separate alteration. An intervention can hold some of the familiar conditions in place while changing others, and a long-

standing question has to be answered, knowingly or not, each time someone chooses how a particular being is to continue.

§1.2 What Comes Apart

Three considerations were named above: the persistence of the body, the connection of psychological content such as memory and character, and the continuation of activity over time. Four kinds of change separate them. For each it helps to ask what stays in place, what varies, and what question is left.

Interruption. Activity stops and later resumes, while whatever carried it is kept intact. The clearest present cases are artificial. Consider a running process that is halted while its state remains on the same storage; later, operation resumes through the preserved processing-and-storage arrangement. None of the system's content has been lost and none of its hardware exchanged. Only the run of activity has a break in it. The artificial case puts the question cleanly. Does persistence require that activity never stop, or can the very same system take up again after a stop? And if it can, what separates that from an outcome that looks the same from outside, in which the stored state is copied to another machine and started there?

Impairment. What a being has retained is damaged while its body lives and its activity continues. Progressive dementia is the familiar case: memories become unreachable, the connections among them weaken, and character may change, across years in which the person is awake, responsive, and cared for as the same patient. Corruption of stored data is the artificial counterpart. Here the body and the ongoing activity are in place, and the connection of psychological content is what gives way. The question is how to describe the result. Has the earlier being been gradually replaced by another, or is this one being, diminished? And if the second, what remains the same while so much is lost?

Replacement. What a being is made of is exchanged while it goes on operating. In a weak form this is as old as life, since a body exchanges its matter continuously. In a stronger form it is current practice: implants and prostheses take over the work of lost or damaged parts and are integrated, more or less closely, into the body's functioning. In the strongest form it is projected: neural tissue replaced part by part with functionally integrated components, possibly until none of the original tissue remains. In this projected case, activity is stipulated to continue and memory and character to carry on undisturbed. What varies is the composition of the body and, at the limit, the kind of thing it is made of. If the being persists through the whole sequence, then persistence does not require keeping the original tissue. If it does not, there should be some point at which it ceased, and something should explain why that point and no other.

Duplication. A being's structure is read, and a second system is produced from the reading. For artificial systems this too is routine: systems are cloned, and systems lost to failure are rebuilt from backups. For living beings it is the projected path of brain emulation, with the original either destroyed in the scanning or left intact. The copy has, by stipulation, the memories and character of the original, as exactly as one cares to specify. Its body is a different one, and its activity begins when it is started. Where the original survives, the difficulty is easy to see. Two beings now carry the earlier

one's psychological content in full, and they are two. If there is a difference between continuing and being copied, it does not lie in the content, which the two share.

Table 1.1. Four ways the familiar conditions of persistence come apart.

Kind of change	Stays in place	Varies	Question left open
Interruption	What carried the activity; retained content	The unbroken run of activity	Can the same being resume after a stop, and what separates resuming from starting a copy?
Impairment	The body; ongoing activity	The connection of memory and character	One being diminished, or one being replaced by another?
Replacement	Ongoing activity; memory and character	The composition of the body, up to the kind of substrate	Does the being persist throughout, and if not, where does it end and why there?
Duplication	Memory and character, as exactly as specified	The body; the origin of the activity	If continuing differs from being copied, in what does the difference lie?

Two remarks on these cases. The first concerns how near they are. The computational examples make the distinctions concrete without requiring the projected biological procedures. In the specified comparisons, one can identify what was kept, what was stopped, and what was produced anew. For living beings, impairment is old, replacement has begun in partial forms, and the rest is projected. The projected cases are used for their structure: they show which conditions a procedure of that kind would hold in place and which it would vary, whatever its timetable turns out to be.

The second remark concerns the three considerations themselves. They are the terms in which persistence has ordinarily been tracked, and the cases show that they can be varied independently. It does not follow that the right account of persistence will be stated in those terms. An unbroken run of activity, for example, may be one way in which a being's continuation shows itself without being what that continuation consists in. Interruption puts exactly that in question.

§1.3 The Question of Existence

None of this would be pressing if the question could wait. It cannot, because it is answered whether or not anyone asks it. Every procedure and every built system embodies a position on the matters its designers did not raise. A pipeline whose stages are scan, model, and run has taken a position on what persistence requires, by treating the matter as one that does not arise. A recovery policy that rebuilds a lost system from its last backup and gives it the old name has taken another. Where theory is silent the default speaks, and a default is a decision no one argued for, made durable by being built.

It helps to be clear about which question this is. Locke observed that sameness is asked of different things under different ideas: whether this is the same mass of matter, the same living plant or animal, and the same person are distinct questions, and their answers need not coincide (*Essay* II.xxvii). The present question is closest to the second, extended beyond the living. It asks about the persistence of a being understood as an organized system that processes and retains, whatever it is made of. It is not a question about sameness of matter, and it is not yet a question about personhood or standing.

Stated in full, the question has three parts. Under what conditions is a later state the next state of an earlier being, and not merely a state that resembles it or was caused by it? How does the continuation of the same being differ from the continuation of similar content and function in another? And how is a stop in a being's operation to be told apart from its end?

These cannot be answered by looking at change over time alone. Replacement and implanted devices raise a question about a being at a single time: what belongs to it, and what merely supports or surrounds it? Artificial systems raise the same question in another form, since their parts may be spread across machines and shared with other systems: what makes these parts one being, and not several or none? An account of persistence therefore needs two components that fit together. One says what makes something a single being at a time. The other says what makes a later state the next state of that being.

The cases also indicate what an answer must be able to do. It has to apply to living, artificial, and hybrid systems in the same terms, since the cases cross those lines and some of the beings in them are already mixed. It has to rest on how a system is organized and how its states come about, because conviction and outward behavior cannot decide the matter: a copy is as certain of its past as the original, and as convincing to everyone else. It has to keep apart two questions that impairment tends to run together, namely whether this is still the same being and how much of that being remains intact. And it has to be usable, so that of a proposed procedure or architecture one can say what it would preserve and what it would end.

§2 sets out URNAVA's response. The requirements just listed do not dictate it, and other responses to the same question are possible. What follows is one proposal: an account of what a being is at a time and of how it continues through time, given in three principles and the way they depend on one another.

§2 — The Three Core Principles

§2.0 The CSC Structure

Take a being while it is in operation: a person in conversation, or a computational system at work on a task. Three things can be distinguished in what is going on. There is processing, the ongoing activity in which each state gives rise to the next. There is a retained structure: what earlier processing has laid down and kept, such as memories, learned associations, and skills, or in a computational system its stored state and learned parameters. And there is a relation between the two that runs in both directions. The retained structure shapes the processing, since what the being does next depends on what it has kept. The results of processing are taken into the structure, which is altered by them and goes on to shape what follows.

The processing, its retained structure, and the relation between them, taken together, are an *arrangement*. URNAVA's three principles describe this one operation from three sides.

Table 2.1. Three aspects of one processing–retention relation.

Principle	Focus	Distinction it explains
Continuity	The route by which later states arise through the arrangement.	The earlier being's next state versus a separately formed successor.
Stability	The cohesion of the historically retained information structure.	Preservation through change versus loss or another realization of the pattern.
Capability	The two-way relation in which structure shapes processing and receives its results.	Constitutive participation versus contact or support.

These aspects, together called CSC, are described through one another. Processing draws on retained structure and changes it; the structure carries those changes into later processing. They are not three components placed side by side.

§1 ended with two questions: what makes something a single being at a time, and what makes a later state the next state of that being. For a being in operation, the proposal is that one relation answers both. What makes it a single being is that its processing and its retained structure form one another in the way just described. What makes a later state its next state is that the state was formed through that same operation, by the arrangement that constitutes the being. What makes it one is also what carries it forward.

This is a description of a being at work. The sections that follow take the three sides in turn and ask what happens when the operation is disturbed: when processing stops, when the structure is damaged or its material replaced, and when the relation between the two is altered.

§2.1 Continuity: The Chain of Self-Succession

Continuity is a relation among the states of a being. A later state succeeds an earlier one, in the sense intended here, when it is formed from the earlier state through the being's own arrangement: the being's processing, shaped by its retained structure, produces the later state, and the results are taken into that structure. A being's history is a chain of such successions. The term *self-succession* marks the fact that the being, as it is constituted at each moment, does the forming of what it is at the next. What belongs to an arrangement at a given stage is a question about the two-way relation described in §2.3; how that arrangement forms a later state is the question here.

Two other relations have to be kept apart from this one. The first is resemblance. A being may come to differ greatly from its earlier self, as an adult differs from the child she was, and two distinct beings may resemble each other as closely as one likes. The second is causal influence in general. Beings are acted on from outside all the time. A teacher's words bring about states in a student; a meal becomes part of a body; a surgeon's repair restores a function. None of this interrupts succession, because in each case what arrives from outside enters an arrangement that goes on forming its own next state. The words are understood in light of what the student already knows, and what she learns is laid into her memory. The repaired tissue is taken up into the body's functioning. What comes from outside can help form the later state. It does not replace the arrangement through which that state is formed.

The same holds when what arrives is a new part. A replacement component comes to belong to a being when the arrangement takes it up: its activity is shaped by the retained structure, its results enter that structure, and it then takes part in the changes that follow, including the taking up of later parts. Incorporation of this kind can in principle run through every part the being began with. At each step, the arrangement that does the incorporating is the one the previous steps produced.

Call the processing–retention arrangement in place at a given stage the *standing arrangement*. Its *organizing relations* are the links through which retained structure is reached, results are bound into it, and changes enter further operation. In incorporation, these relations receive and organize the incoming part; the resulting arrangement then receives later changes. “Standing” names this role in the sequence, not an unchanging set of components. The organizing relations may themselves be revised through that sequence.

Why should this relation, and neither resemblance nor causal connection, decide whether a later state belongs to the same being? The reason lies in the ordinary case. Where the familiar conditions travel together, we count the adult as the child grown, through every change of matter, belief, and character. What explains that judgment is not likeness, since the two are unlike. Nor is it merely that the child's states caused the adult's, since the child's states also caused states in her parents and friends. It is that every difference between child and adult was produced through one arrangement and absorbed into it. Each change was shaped by what had been retained so far, and became part of what shaped the next. The operation that constitutes the being is the same operation that produces its changes and keeps them. Succession is therefore no special criterion devised for hard cases. It is what was doing the explanatory work in the easy ones, where the conditions that accompany it kept it out of view.

If that is right, the difference between a being's changing and the formation of a separate being can be stated. A being changes when its later state is formed through its own arrangement, however much help, material, or information arrives from outside. A separate being is formed when something else sets up an arrangement: another being, or an apparatus working from a description. This can happen with the first being's full support and out of its materials, and the end states may match in every respect. What differs is what did the forming, which is a fact about history that no inspection of the end states reveals.

Parfit's discussion of psychological continuity brings out part of this. He distinguishes psychological connectedness and continuity from numerical identity and considers what causal conditions the psychological relation must meet (Parfit 1984, Part Three). These are questions about the relation between earlier and later stages, rather than resemblance alone. CSC's answer to the question of persistence is the formative succession just described. That relation is a specific form of what the metaphysics of persistence calls immanent causation, in which a thing's later states are brought about by its own earlier states, as distinct from causation passing from one thing to another (for a detailed treatment see Zimmerman 1997). The general distinction does not settle which cases are which. The work lies in saying what a being's own arrangement is, how succession passes through a stop, and how it differs from help that arrives from outside.

Processing can stop. The clearest present cases, as §1 noted, are computational, and three of them can be set side by side. In the first, a process is halted while its state is kept in the arrangement that was running it, and later processing starts again in that preserved arrangement. In the second, the state is read out and used to set up a separate arrangement, which is then started. In the third, the process ends and nothing of its state is kept; later a process of the same name is started from defaults.

In the first case the processing really did stop, and nothing was processed during the interval. What was preserved is the retained structure, together with the rest of the arrangement in which processing had been taking place. When processing begins again, its first state is formed from that structure. It takes up what the earlier processing had laid down and the arrangement had kept, through the relation that held while the system ran. The stage after the stop therefore stands to the stage before it as successive stages stand to one another during operation. Whether the two stages belong to one history is a question about how the later stage was formed. It is a different question from whether processing went on in between, to which the answer here is no.

In the second case, the first state of the new arrangement is formed by whatever did the setting up, working from a description. The earlier arrangement supplied the description and took no further part. This is so wherever the new arrangement is placed. Suppose the state is read out, the machine is cleared, and a fresh arrangement is set up on it from the reading and given the old name. Storage device, machine, and name are all the same, and the case is still of the second kind. None of the three is what makes a later state the next state of an earlier being.

It may be objected that the first case involves reading too, since a restarted process must read its stored state before it can go on. It does, and so does every moment of ordinary operation, in which retained structure is drawn on continually. Reading as such is not the difference. In the first case the reading is the arrangement's own way of drawing on its retained structure, the same by which that structure shaped processing before the stop. In the second, the reading yields a description that

something else uses to build. What belongs to an arrangement, and what serves it from outside, is the subject of §2.3. The present contrast needs only the difference between an arrangement taking up its own retained structure and a separate arrangement being set up from a description of it.

In the third case nothing is kept from which later processing could be formed. The new process has a predecessor in name only.

Two terms mark the contrast. An *interruption* is a stop in processing after which succession resumes through the retained structure. *Severance* is the failure of succession between an earlier being and a candidate later state, because that state was not formed through the earlier being's arrangement. So defined, severance is a fact about the candidate. It is not something that befalls the earlier being. When a being is copied and goes on operating, its own arrangement continues to form its next states as before, and being read is one more thing that has happened to it. The copy's first state fails to be the original's next state, and nothing follows from that about what becomes of the original. If the original is destroyed in the copying, that is a further fact about its structure, to be established on its own grounds.

The relation leaves one kind of case open. Nothing said so far excludes an arrangement's dividing in such a way that each part carries on as a complete arrangement, forming its next states from its share of the retained structure, or two arrangements' merging into one. In these cases, later states may be formed through earlier arrangements rather than from descriptions of them, so the contrast with copying does not settle the whole family. What should be said depends both on the formation route and on what makes an arrangement a single unit. §2.3 takes up the question of units, and §3 examines the cases.

The account of the first case relied on a phrase not yet examined: the retained structure, kept through the interval. If succession passes through a stop only by way of that structure, it matters what it is for a structure to be kept, and what distinguishes the structure that was kept from another exactly like it.

§2.2 Stability: The Retained Structure

What a being retains is not a heap of items. Information is held in relations: associations among memories, strengths of connection, the indices and contexts that tie one trace to others and make it reachable. Something learned is retained by having a place in such a web, from which it can be reached and brought to bear; an inscription that nothing connects to anything is not yet a memory. The same holds for a computational system, whose stored values are organized by the addresses, keys, and learned parameters through which they are used. Stability is the holding together of this relational structure as a single base. It is not the quantity retained, the constancy of what is retained, or the sameness of the material that carries it.

The last of these points has a long history. Locke distinguished the identity of a mass of matter, which is the same only while it consists of the same particles, from that of a living plant or animal, which remains the same while its particles are exchanged, so long as the new particles are united in one continued organization and share in one life (*Essay* II.xxvii). Maturana and Varela made a related point in general form: what defines a system as the unity it is lies in the relations among its

components, and the same relations can be realized by different components (Maturana and Varela 1980). CSC takes from these the thought that an organized thing is to be followed through its organization and not through its material. It applies that thought to one structure in particular, the structure in which information is retained. It does not claim that every self-maintaining organization is a case of Stability. Nor does it require what autopoiesis requires, that a system produce its own components: computational systems retain without producing the hardware that carries what they retain.

A constructed computational case makes the same point. Let a store keep its contents from fading by periodically rewriting its cells, and let it move entries from worn cells to fresh ones as part of the arrangement's own upkeep. The operations that maintain its indices and links preserve each entry's place in the structure through these moves. Over time every physical carrier is exchanged. Although each local move reads and writes a value, it maintains the existing store rather than initializing a separate arrangement from a description of it. The carriers change; the retained relations are maintained through the arrangement to which they belong.

A stable structure is not a fixed one. It learns, which adds relations; it forgets, which weakens or removes them; it consolidates what it holds, and it is repaired. These are changes within the structure. New content is bound into relations already standing, losses are losses from it, and a repair is taken up by what surrounds the repaired part. Because a structure can hold together more or less well, Stability comes in degrees. In progressive dementia, as §1 described it, connections weaken, contexts detach, and regions of what was retained fall out of reach. Corruption of stored data is the computational counterpart.

Here the two questions that §1 asked to be kept apart can be separated. The cohesion of its retained structure is one aspect of how much of a being remains intact, and admits of degree. Whether it is the same being is a matter of how its states have come about. Where the states of a person with dementia continue to be formed through one arrangement, progressively damaged, each stage of the damage is a change to the structure that was there before. On this account the person remains the same being through those changes, diminished, and the loss is a loss of what that being retains. The separation has a limit. Succession runs through a retained structure, and if nothing of the structure remained there would be nothing from which later states could be formed as the being's own. Degrees of damage do not translate into degrees of identity; the complete loss of the structure is a different matter.

A structure can be described, and a description can be realized more than once. So there are two things one may mean by "the same structure." One is the type: the pattern of relations that a complete description would capture. The other is the particular structure that a being's processing has laid down and that has been kept since, with whatever changes it has undergone along the way. Two exact realizations of one description are the same in the first sense and two in the second. It is the particular, historically continued structure that matters for succession. When processing resumes after a stop, its first state belongs to the earlier being's history because it is formed from what that being's processing actually laid down and what was actually kept, and not from another realization of the same pattern. For a structure to be kept, in this sense, is for its relations at each time to be there because they were there before, maintained or modified through the standing arrangement. Its

material may be exchanged, and it may be moved whole from one place to another. What it cannot be, and still be the structure that was kept, is set up again from a description.

This gives the means to compare two uses of a record. Suppose a structure is damaged and a record of its earlier condition exists: a scan, a log, a backup. The record may serve as a guide to repair. It shows where the damage lies and what belongs there, and it may even supply the content of a part that was lost. But the mending is done on the standing structure. The relations that remain determine where the restored part fits, the part is bound into them, and what results is the earlier structure, mended. Alternatively, the record may serve as a specification from which a new structure is set up. Then the record is the source of the arrangement, and the damaged structure, if it still exists, takes no part in the building. The result realizes the same type, and it is another structure.

The difference between the two is not the amount replaced. A long enough series of repairs could in principle renew every part, as incorporation can renew every part of a being in operation. Nor is it the presence of original material or sameness of place, since material can be exchanged and a structure can be moved. It is the part the retained structure plays in forming what comes after: whether it is what gets mended and built into, or only what gets described. The difference can be made visible by letting the damaged original survive beside a structure newly built from its record. There are now two, and only one of them is the structure that was damaged. Mending that one and building the other were different acts all along, whatever becomes of the original afterward.

Not every case falls cleanly on one side. Where damage is severe and most of what is restored comes from a record, it can be hard to say whether what remained was still receiving the repair or had become a fragment attached to a rebuilding. Some portions may have been bound into standing relations while others were set up independently. The account does not make such cases easy. It says what the difficulty is about: the role of what was retained in the formation of what came after.

One thing has been assumed throughout. A retained structure, however cohesive and however faithfully kept, is not by itself a being. A library retains, and so does an archive on which nothing has ever run. What makes a retained structure part of a being is that it takes part in processing, shaping it and receiving its results; and what made the restart in §2.1 a resumption was that processing took up the structure again through that same relation. What is that relation, and what makes processing and structure a single unit and not two things in contact?

§2.3 Capability: Mutual Legibility

A program can query a database it never writes to. A reader can spend a lifetime consulting a library without adding to or revising its holdings. In both cases a processing activity and a retained structure are in contact, and information passes from one to the other. What is missing is what §2.0 described: the results of this processing do not return to the consulted structure and enter the relations that guide its later use. The third principle concerns the relation that is present in a being and absent here. URNAV calls it Capability, and the name needs a caution at once. It does not refer to intelligence, performance, or the difficulty of the tasks a being can manage. It refers to the capacity of processing and retained structure to take each other up, which is better described as their mutual legibility. A

very simple arrangement can have it in full, and a powerful processor that draws on a fixed store and returns nothing to it lacks it.

The relation has two directions, and each is an operation that can be described. In the read direction, retained relations are used in the forming of what processing does next. They serve as premises, as the material operated on, and as what sets the following transition: a learned association determines what comes to mind, and a stored parameter determines how an input is transformed. In the write direction, a result of processing is bound into the relations already standing. It is indexed, weighted, and linked to its context, so that it can be reached later and can take part in later processing. Output that is merely left on a storage device does not amount to this. A log that is appended to and never drawn on records what happened without becoming part of what shapes what happens next. Nor should the storage be pictured as a second agent that reads and judges. The binding is done by operations that belong to the arrangement's own processing: the routines that update an index, or in a brain the processes that consolidate what has been experienced.

A constructed case shows the two directions working together. Let a system work through a series of tasks, keeping a store of past cases. Each entry links a kind of situation to what was tried and how it went; entries are indexed by features of the situation and carry weights that say how far each should be trusted. A new task arrives. Processing retrieves the entries whose features match, and their weights settle which approach is attempted. That is the read direction: what was retained has set the next transition. The attempt fails for a reason the system can identify. A binding operation, part of the system's own processing, adds an entry for this outcome, links it to the entries that led to the attempt, lowers their weights, and extends the index so that the new entry will be found under the relevant features. That is the write direction. Later a task with overlapping features arrives. Retrieval now brings up the new entry, the lowered weights steer processing toward a different approach, and the result of that attempt is bound in its turn. One result has entered the retained relations, has been used in later processing, and through that processing has led to a further change in the structure.

A *cycle* is one passage from retained structure through processing and back into retention. *Coupling* names the standing two-way relation that makes such cycles possible; *mutual legibility* names the capacity of processing and structure to take part in that relation.

The case also shows why this relation is a separate matter from the cohesion of the structure. Suppose the processing component is revised, and after the revision it expects the index in a different form. The store is untouched and as cohesive as it ever was. But retrieval now returns nothing usable, processing proceeds as if from nothing, and whatever it records goes into a new store to which the old one is not linked. The structure stands and the processing runs, and neither any longer forms the other. Now suppose instead that retrieval works as before while the binding operation fails for one class of tasks. Old entries are found and used, so the past goes on shaping what the system does. New results from those tasks are not indexed or linked; they are lost, or pile up where nothing will find them. Results from other tasks are still bound as before. For the affected tasks, the system has no newly bound outcomes to draw on. In the first variant the read direction has failed while the structure is intact. In the second the write direction has failed for that class while the structure is intact and still in use. Neither is a loss of cohesion, and neither is a stop in processing.

Both directions can weaken by degrees, and to different extents. Part of a structure may fall out of reach while the rest is used as before; some kinds of result may fail to bind while others bind normally. Three things should therefore be kept apart. One is how far the coupling extends: how much of the structure takes part in processing, and how much of what is processed is bound. A second is whether something takes part in the relation at all, which is the question of what belongs to the being. A third is the failure of the relation as a whole. It would be a mistake to run these together in either direction. Two people in conversation alter what each other retains, and they do not thereby become one being, so not every two-way influence makes a unit. And a partial failure in one direction leaves a being impaired, as the second variant shows; it is not an ending.

Where the relation fails as a whole, so that processing and structure are both present and neither forms the other, URNAVAspeaks of *estrangement*. The first variant, taken to its limit, is a case. Estrangement is a failure of mutual legibility, and it is not by that fact final. What matters is how the coupling is restored, and here the distinction of §2.2 between mending and rebuilding applies again. In one kind of recovery the components that were kept are mended to restore this coupling: an adapter is built into the arrangement, the revised processing can once more use the old index, and its results are once more bound into the old store. The structure that comes back into use is the structure that was kept. That the adapter was designed and installed by engineers does not alter this, any more than a surgeon's help turns healing into replacement. In the other kind, the old store is read, a new store is set up from the reading in the form the revised processing expects, and the old one is set aside. What processing now draws on is a structure built from a description of the earlier one. By the argument of §2.2 it is another structure, and the one that was kept has not come back into operation, however well its successor serves.

Questions about what belongs to a being are questions about this relation, and position does not answer them. A component inside a skull or a casing may do no more than deliver signals that processing treats as input from elsewhere. A store on a distant machine may be what a system's processing draws on and binds its results into. Nor does dependence answer them. A power supply, a cooling system, or a supply of nutrients may be indispensable without participating in this two-way relation. A service, too, can compute something on request and return an answer that the being treats as an input from elsewhere. Help of these kinds differs from participation, in the way §2.1 distinguished what helps to form a later state from the arrangement through which it is formed. The relation gives the terms in which such questions are asked. It is not yet a procedure for deciding every case.

Clark and Chalmers argue that a resource's location outside the body does not exclude it from cognition if it plays the relevant cognitive role. In their example, Otto's notebook serves as memory; his access by perception does not, they argue, mark a decisive difference from biological recall (Clark and Chalmers 1998). In discussing his standing beliefs, they emphasize constancy, easy availability, and automatic endorsement. CSC shares the negative point: neither skin and skull nor a device's casing settles what belongs to a being. Its positive question differs. Rather than asking whether a resource plays the role of a belief or cognitive process, CSC asks whether it participates in the two-way formation of processing and retained structure that constitutes a being and carries its history.

These are different grounds for inclusion. Whether perceptual and interpretive access can meet the latter condition is the question of directness.

Open definition — Directness. Whether participation requires direct use of retained states, excluding access through an interpreted stand-in, or can include mediated access integrated into the read–bind–revision cycle remains undecided. Neither treatment is adopted here.

The order of explanation matters here. The relation has been described as holding between processing and a retained structure, without first assuming a being to which both belong. The unit comes second: a being is what the relation binds together. To say that processing takes up retained structure “as its own” is therefore not to report a feeling of ownership or an awareness of self, and it is not to appeal to a prior fact about what the system includes. It is to say that the structure functions in the processing in the ways described, and that the results of the processing are bound into it. This follows the point taken from Maturana and Varela in §2.2, that a unity is defined by relations among components and not by a list of them. The relations differ. Theirs are relations of production, by which a living system makes and replaces its own components. The relations at issue here are those of retention and mutual use, which a system can have without producing the parts that carry them.

How far a unit extends is a further question, bounded by two errors. One would merge every pair of arrangements that affect each other in both directions. The other would count as a being every small cycle that can be traced within an arrangement, such as a value that a routine reads and rewrites. Between them lie the cases that matter: arrangements that come to draw on and bind into a common structure, and arrangements within which narrower cycles can be distinguished.

Open definition — Maximality. Whether only the largest arrangement closed under the relation counts as a being, or distinct completed relations can constitute beings at more than one scale, remains undecided. §3.3 states where the alternatives change a judgment.

The harder cases of joining, nesting, and division are taken up in §3. What remains for §2 is to say how the three principles work together.

§2.4 The Intersection: One Structure, Three Principles

The three principles have been presented one after another, but they are used together, in a single explanation. Take the restart of §2.1. To say that the stage after the stop is the next stage of the same being, three things had to be said. The later state was formed through the earlier arrangement and not set up separately from its description: that is Continuity. What it was formed from was the structure actually laid down and kept, and not another realization of its pattern: that is Stability. And processing took that structure up again in the relation that had held before, so that there was one arrangement to resume and not a processor and an archive side by side: that is Capability. Each statement uses terms that the others explain. Continuity speaks of the being’s own arrangement, and what makes an arrangement a unit is given by Capability. It speaks of succession passing through a

stop, and what passes through is given by Stability. Capability in turn describes a relation at a time; whether that relation carries on as the history of one structure is again a matter of Continuity.

Mutual dependence of this kind does not yet show that each is needed. That is shown by what cannot be said when one is left out. Leave out Continuity, and compare the system that resumed with an exact copy started on another machine. Both have a cohesive structure, and in both processing and structure form one another. Their current structural pattern and two-way coupling can match. What distinguishes the one that resumed is how its present state was formed, and only succession states that. Leave out Stability, and there is nothing to say about what carried the being through the stop, no difference between a structure mended and a structure rebuilt from a record, and no account of diminution through loss of structural cohesion. Leave out Capability, and the estranged system of §2.3 cannot be told from a working one. Its processing continues, each state formed from the last; its store is kept, as cohesive as ever; and neither forms the other. Without the third principle there is also no account of what belongs to a being, so that the phrase “its own arrangement,” on which the account of succession relied, would have no content.

The principles distinguish aspects of one operation. A structure may be kept while processing stops; cohesion may decline along a continuing history; and mutual legibility may fail while structure and processing remain. These contrasts show why the aspects must be distinguished, but do not establish that any one can hold in full while another is wholly absent. Their combined role is to explain the differences that §1 brought into view.

Putting constitution and persistence together has several consequences. The first concerns beginnings. A system set up from a copy is not the next stage of the original, and that is all that severance says about it. It says nothing against the new system’s being constituted in its own right. If its processing and its structure form one another, it is a being, with a history that runs forward from its first operation. What it carries over from the original is an inheritance and not a past of its own. This also removes an apparent regress. §2.0 described the retained structure as what a being’s processing has laid down, and it may be asked how any being could then begin. It begins when processing and a structure first enter the two-way relation. The structure may have been grown, built, or copied by something else. From the first cycle onward it is what this being’s processing is making of it.

The second concerns damage. How much of a being remains intact is not measured by Stability alone. It depends on how well the retained structure holds together, and also on how much of that structure still takes part in processing and how much of what is processed is still bound. None of these degrees is a degree of identity. But the complete loss of what constitution rests on, the structure or the coupling beyond any mending, is a different matter from any degree of impairment.

The third concerns stops. The stages before and after a stop can belong to one history, on the conditions given in §§2.1 and 2.2: the structure is kept, and the arrangement through which it was formed is what takes it up again. What those conditions require is the continued preservation of the retained structure and of a route by which the arrangement can resume, including through repairs integrated into what remains. Whether processing ran during the stop is, as before, a separate question. This account of succession between the stages leaves undecided whether the being exists throughout the interval.

The fourth concerns failures and endings, which the account keeps apart. A failure of succession concerns a candidate later state and does not by itself establish what has happened to the earlier being. A failure of coupling impairs a being or halts its operation, and may be repaired. Neither failure by itself establishes an ending. A being has ended when no retained structure and route remain through which later states could be formed as its own: when the structure is lost, when what remains permits only the separate construction of another arrangement from a description, or when the earlier coupling has been dissolved into other arrangements without preserving a route for its own resumption.

These distinctions give the questions to be asked of any actual transformation. What remained? How was it changed? And from what structure, through what relation, was the later operation formed? §3 puts these questions to cases that already occur, to technologies that are projected, and to cases in which beings are joined or divided.

§3 — Application and Judgment

§2 ended with three questions to put to any transformation. What remained? How was it changed? And from what structure, through what relation, was the later operation formed? §3 asks them of cases, arranged in three tiers. The first takes changes that already occur, in people and in computational systems. The second takes technologies that are projected, described by their conditions and not by a timetable. The third takes cases in which beings are joined or divided. The order is one of widening application, from what is actual to what could be constructed, and nothing in it predicts that the later tiers will arrive. The aim in each case is to say what continues, what begins, and why. The comparisons concern the specified histories; a procedure's name alone does not determine the judgment.

§3.1 Tier I — Present Cases: Interruption, Impairment, and Recovery

A change in activity does not by itself show that a being has ended. The cases below concern losses and changes to the arrangement: impairment, dependence on help, and recovery through what remains.

Impairment

In progressive dementia, what a person has retained comes apart by stages. What matters for the present question is how each later condition arises. A woman who no longer recognizes her street may still hum the songs of her youth, reach for her husband's hand, and bristle at being hurried, as she always did. What remains of her retained structure is still shaping what she does. What she undergoes now, the damage included, is a change to that structure and not the installation of another. Her later states are formed through the one arrangement, progressively diminished, and that is the account's reason for saying that she is the same being. It is also why the loss is hers. What has happened is not the arrival of a stranger. It is something undergone by the person who was there all along, and that is what makes it grievous.

Impairment does not always take this form. Access to what was retained and the binding of new results can fail separately, as the variants of §2.3 showed. In the case of patient H.M., severe difficulty forming lasting memories of new events coexisted with access to much earlier knowledge. He also learned and retained a new motor skill over several days of practice without recollecting the practice (Squire 2009). In CSC terms, some results were still being bound into what he retained while others were not. The case is therefore not one in which a principle has simply switched off. The coupling between processing and retained structure has narrowed, in one direction and for some kinds of result, in a being whose states continue to be formed through the arrangement that was there before.

Olson's animalism reaches a similar verdict about such cases by another route. On his view each of us is a biological organism, and no psychological relation is necessary or sufficient for an organism to persist; a person could survive the loss of all psychological content so long as the organism remains alive (Olson 1997). CSC likewise does not make psychological continuity the condition of persistence: a loss or alteration of memory and character is not, by itself, the replacement of one being by another. Its reason differs. The person continues because her later states are formed through the arrangement that formed her earlier ones, and not because a biological life continues as such. In the cases of this section the two reasons pick out the same being. They come apart where what carries the arrangement is no longer a living body, which is a matter for §3.2.

Help, Repair, and Ending

A being may come to depend on outside help for the conditions of its operation, and the dependence may be permanent. That is not an ending, and it is not a replacement. §2.3 distinguished support from participation: a device can supply conditions or perform a task for a body without thereby joining the processing–retention relation that constitutes the being. Repair goes further than support, since it changes the arrangement, and here the question is again what the remaining relations do. A person relearning after a brain injury is helped by therapists, by family, by photographs and notes. What she is told and shown does not assemble a second person. It is understood through relations that survived and is bound into them, and where a connection has been lost, the surviving ones settle what can be rebuilt and where it attaches. The help may be extensive and the records elaborate. The repair is still hers, because what remained of her is what received it.

The same distinction separates a being that cannot at present be restored from a being that has ended. §2.4 described an ending as the loss of any retained route through which later states could be formed as the being's own. Whether that condition has been reached is a question about the being, and not about the skills of those attending it. The point is clearest in a computational case. One system has halted because a part has failed that no one now knows how to replace; its store is intact, and the rest of its arrangement is preserved. In another, the store contained all of the retained structure, and its degradation has left neither those relations nor a route through another preserved component. The first cannot be resumed today, although a retained route remains. In the second, the loss has removed the structure through which its own succession could resume. Which description fits a given clinical situation is a question of fact about what remains, and the name of a diagnosis does not settle it.

Recovery in Computational Systems

Computational cases let the differences be specified without attributing them to a particular product. Three contrasts separate recovery of a service, recovery of a coupling, and continuation of the being that provided the service.

Table 3.1. Three recovery contrasts. Judgments apply to the conditions described below.

Comparison	First case	Contrasting case	Decisive difference
Resumption / backup rebuilding	Processing resumes through the preserved arrangement: the earlier history continues.	A separate arrangement is initialized from a record: content is inherited.	Preserved succession or separate formation, not the service name.
Partial mismatch / estrangement	Some retained relations remain in use: the being continues impaired.	The earlier processing–store coupling fails as a whole; that failure alone does not establish an ending.	Which retained relations still participate, including any carried by the processor.
Repair / replacement store	The kept store returns to use through a repaired correspondence, if the adapter belongs to the coupling.	An export specifies a separate store while the earlier store is set aside.	The preserved structure’s role in recovery; adapter membership remains a further condition.

Resumption and backup rebuilding.

A system is halted with its store and the rest of its arrangement preserved, and later processing begins again through that arrangement. A second system is lost, and a replacement is initialized from last night’s backup and given the same name and network address. Let the backup capture all the data relevant to the comparison. From the users’ side the recoveries can look alike: the service is back, with the same data and behavior.

In the first, later states are formed through the preserved arrangement. In the second, a new arrangement is set up from a record. If its processing and structure form one another, it is a being in its own right, with inherited content rather than the earlier being’s history. Restoring a service and continuing the being that provided it can come apart. An operator whose aim includes the second needs to know which result the recovery plan delivers. Every resumption reads stored state, and a command called “restart” may resume a preserved arrangement or build a fresh one. Neither reading nor the command name decides which route was taken.

Partial mismatch and estrangement.

After a revision to its processing component, a system can no longer use one region of its store, while the rest is reached and bound into as before. This is impairment: one being, with part of what it retained out of reach. Now let the revision cut processing off from the whole store, so that new results enter a fresh store unlinked to the old one. The earlier coupling has failed, and a new processing–store coupling has begun. The latter does not restore the former. Nor does the earlier coupling’s failure establish an ending while the old store remains available for repair.

The running system’s history requires one further fact. If the processing component itself carries retained relations from earlier activity, part of the earlier structure is still at work, and the being can continue impaired. If everything retained lay in the disconnected store, the new operation draws on none of it. Losing access to an entire storage device is therefore not automatically losing access to the being’s entire retained structure.

Repair and a replacement store.

Now compare two remedies, fixing the old store as the entire retained structure relevant to this contrast. In one, the mismatch is traced and an adapter is fitted so that revised processing again retrieves from the kept store and binds results into it. In the other, the old store is exported, a separate store is generated in the expected format, and the old one is retired. The second procedure initializes the new structure independently; it does not incorporate replacements into surviving relations of the old structure.

Both remedies can change format. The difference is that the first brings the kept structure back into use, with its relations determining what the adapter must do; the second uses a description to establish another structure. Their similar output does not erase this difference. The first restores the earlier coupling only if the adapter participates in the arrangement in the sense required by §2.3. If it merely consults the old store and delivers answers from outside, it supplies a different route of access. The membership condition remains unresolved for this adapter case; format conversion alone answers neither way. The second remedy establishes a new coupling with inherited content, leaving the earlier structure out of operation.

§3.2 Tier II — Future Technologies: Replacement, Preservation, and Transfer

The biological transformations below are projected cases, specified by their conditions rather than a timetable. The neural-interface study cited in §1 decoded signals from living tissue into text (Willett et al. 2023); it did not perform the wholesale replacement considered here. The accompanying computational comparisons isolate formation routes without assuming that the biological procedures are available.

Gradual Replacement

Suppose that neural tissue could be replaced, a small region at a time, by artificial components. Each component is built so that, once installed, it does within the arrangement what the tissue it replaces did: its activity is shaped by what the being has retained, it carries its share of the retained relations, and what passes through it is bound into the structure that shapes later processing. Installation is followed by a period in which the component is adjusted through the being's own activity, as new tissue would be. Later replacements are made into an arrangement that already includes the earlier ones, and the earlier components take part in receiving them. Carried far enough, the sequence ends with no original tissue left.

On the account of §2 the being persists through the whole sequence. At every step the later state is formed through the arrangement as it then stands. Each component enters by incorporation, as §2.1 described, and the retained structure is kept throughout, its material exchanged while its relations are maintained and modified through the standing arrangement. The comparison extends Locke's distinction between a living thing and its constituent matter to a further, CSC-specific proposal about changes in substrate. There is no step at which something else sets up an arrangement from a description, and so no step at which a separate being is formed.

It is tempting to credit this verdict to the gradualness of the procedure, and that would be a mistake. Suppose a second device is built over the same years and at the same pace. Its components are made

one by one to match regions of the person's brain as successive scans report them, and they are assembled on a bench. None is ever installed. When the device is complete it is started. This procedure is as slow as the first, it never interrupts the person's activity, and at each stage it adds only a small part. Its product is nonetheless a separate arrangement set up from a description. Its first state is not the next state of the person, who meanwhile continues as before. What separates the two procedures is the route by which the parts enter: into the standing arrangement, to be shaped by it and to take part in what follows, or into something assembled beside it. Neither slowness nor leaving the original's activity uninterrupted supplies the missing incorporation.

The first verdict rests on a stipulation that should be made explicit. The components were described as taking part in the arrangement as the tissue did. A component might instead deliver its states to the rest of the brain only as signals to be interpreted, as an instrument is read, and receive results in the same manner. Whether a component of that kind belongs to the being or serves it from outside is the question of directness left open in §2.3. If such components only serve from outside, continuation would still depend on a remaining arrangement that meets CSC; external help alone would not establish survival through its complete replacement. If the stipulated endpoint no longer sustains biological life, an account that makes such life necessary for persistence would also diverge from CSC. That divergence concerns the continuation of life, not merely the fraction of original tissue left. CSC follows that arrangement because it connects constitution with change in the same terms across living, artificial, and hybrid cases. It explains persistence through losses of memory and character by tracing how the remaining organization carries later states. In the replacement sequence, the same explanation follows each new part into the relations that receive it and incorporate further changes. No step substitutes a separately formed being for the arrangement doing this work. Requiring biological life to continue would add a condition not supplied by that history. The disagreement is whether this additional condition is necessary, rather than where to draw a line in the proportion of tissue left.

Preservation, Transport, and Repair

A second family of projected procedures halts a being's processing altogether and preserves what it has retained, with the aim of later resumption. Three changes may be involved, and they are different changes. The stop has been discussed: if the retained structure is kept and processing later begins again through the arrangement, the stages before and after belong to one history, however long the interval. Moving the preserved arrangement from one place to another changes nothing that the account consults, since a structure can be moved whole. And damage suffered before or during preservation may be repaired. If the repair is made into what was kept, guided by whatever records exist, so that the surviving relations settle what is rebuilt and where it attaches, it is repair in the sense of §2.2, and what resumes is the being that was halted. Sameness of place is not required, and neither is the survival of every original part.

The same technology could be used otherwise. The preserved structure might be read in full and a new arrangement built from the reading, after which the original is discarded. However faithful the result, this is the second case of §2.1 followed by the destruction of the original: a new being begins with an inheritance, and the being that was preserved does not resume. The two uses may

be indistinguishable to anyone who meets the outcome. They are different outcomes for the being concerned, and an undertaking to preserve someone should say which of them it promises.

Transfers between storage media raise the same question at smaller scale, and the name of the operation does not answer it. The two transfer procedures compared here both write information to a new medium and eventually remove it from the old one. What matters is the route. In one kind of transfer a new storage unit is joined to a working arrangement. The arrangement's own operations begin binding new results into it and, over time, re-bind existing entries there, each indexed and linked by the operations that maintain the structure, while processing draws throughout on one structure spanning both units. When the old unit holds nothing, it is removed. This is replacement by incorporation, applied to the store. In another kind the system is halted, its contents are exported, a new system is initialized from the export on other hardware, and the old one is erased. That is separate formation followed by destruction. The first description carries a condition where the new unit differs in form from the old and every access passes through an adapter: it then depends, as in §3.1, on what it takes for an adapter to belong to the arrangement. The second description does not depend on that question.

Severe disassembly is judged in the same way. Suppose a preserved structure must be taken apart to be moved or repaired. What decides the outcome is the part the surviving relations play in putting it together again. If the pieces retain the particular relations through which reconnection proceeds, and each repaired connection enters the structure that guides the next repair, reassembly follows the preserved route described in §2.2. Generic fittings or original material alone do not establish that route. If the pieces are reduced to a stock of material and an arrangement is imposed on them afresh from a plan, the result is a structure built from a description, out of original material. Original material is not what the account follows, and the word “disassembly” covers both.

Table 3.2. Similar transformations, different formation routes.

Transformation	Through the retained arrangement	Through separate formation
Component replacement	Installed parts enter the existing relation and help incorporate later parts.	Matching parts are assembled beside the original and started separately.
Preservation and repair	What remains guides repair and is taken up in resumed processing.	A reading supplies the specification for a new arrangement.
Transfer between media	New storage enters one continuing structure before the old unit is removed.	Export, separate initialization, and removal of the original arrangement.
Disassembly and rejoining	Particular retained relations guide reconnection and receive its results.	A plan imposes an arrangement on a stock of material.

The left-hand route carries the earlier history; the right-hand route forms another arrangement. Cases using adapters also require the membership condition identified in §3.1.

Copies, Uploads, and Archives

Consider two versions of scanning a brain and running a model of it elsewhere. In both, the being's structure is read and a new arrangement is set up from the reading. In one version the reading destroys the original; in the other the original is left intact and goes on operating. The two should be compared with the formation of the copy held fixed, because it is fixed: in each, the copy's first state is set up from a record, by whatever does the building. In neither version, therefore, is the copy the next state of the original. If its processing and structure form one another it is a being, beginning with a rich inheritance. The versions differ in what happens to the original. Where it is left intact, its own arrangement continues to form its next states and it persists, as §2.1 said. Where it is destroyed, it has ended, because of what was done to its structure. The destruction of the original gives the copy no route of succession that it would otherwise lack. It changes how many beings there are afterward, and not which of them continues the earlier one.

Parfit's teletransporter is the best-known form of this pair. A traveler is scanned, and a replica is made at the destination from the record. In the simple case the scanning destroys the traveler; in the branch-line case it does not, and traveler and replica exist together for a time. Parfit used the pair to separate psychological continuity, which the replica has in full, from identity, and to ask which of the two matters in survival (Parfit 1984, Part Three). CSC takes up a different part of the matter. It asks what the replica is and how it came to be, and its answer is the same in both cases: the replica was formed from a record; the traveler's arrangement supplied the record, while a separate procedure built the replica from it. That is a claim about the replica's history, and it is compatible with everything in Parfit's description of the cases. What follows for how a person should regard such a journey is a further question, not pursued here.

Inactivity also leaves three different histories to distinguish. The descriptions below concern what was kept and how it came to be.

Table 3.3. Three inactive configurations and the histories they can support.

Configuration	How it came to be	Relation of later processing to the earlier history
Preserved arrangement	The structure used in earlier operation has been kept.	Processing resumed through it continues that history.
Archive	A record was produced by reading an arrangement.	A separate arrangement built from the record inherits content; it does not resume the source.
Never-run arrangement	A structure was built but has not yet entered a processing–retention coupling.	Its first qualifying operation begins its own history.

The new being may inherit the archive's content and pursue the earlier being's projects. That inheritance does not turn the source's past into its own succession.

One Being in Several Places

The last case of this tier concerns place. Suppose a being's processing is carried out by units in several locations, and its retained structure is held across several stores, joined by channels fast enough that the whole works as the arrangement of §2.3 worked: retained relations across the sites jointly shape processing, whose results are bound into one structure spanning them. This is one coupling realized in many places, and so it is one being. The number of devices does not give the number of beings, any more than the number of cells does. Clark and Chalmers (1998) made the corresponding point about cognition, that location does not settle what belongs to a cognitive process. The present claim concerns what constitutes a being, and it rests on the two-way relation of §2.3 and not on the role a resource plays in guiding action. Whether a given channel between sites sustains that relation, or only passes messages between arrangements that remain separate, is a question about the channel, and where its traffic passes through adapters it inherits the open question already noted.

One caution applies. That a distributed arrangement is a single being now says nothing about how it came to be distributed. A being might reach this condition by incorporation, its arrangement extended site by site while it went on operating. The same final arrangement might be produced by copying a being's structure to several sites and starting it there. The first continues the earlier being and the second begins a new one. What makes something one being at a time and what makes it the next stage of an earlier being remain two questions, here as everywhere. The case also leads to a harder one. A single being spread across several places has so far been assumed to have no parts that are beings themselves. How does that differ from the case in which several beings, each with its own coupling, come to form a larger arrangement together?

§3.3 Tier III — Integration and Beyond

§3.2 ended with a question: how does one being spread across several places differ from several beings that come to form something together? The comparison holds the kinds of components fixed and varies how they are organized. Begin with two arrangements, A and B, each of the kind described in §2.3: each has its own store of cases, draws on it, and binds its results into it. A third, C, is added where needed. One stipulation keeps the question of directness out of the comparison. Wherever a member is said below to draw on a structure or bind into it, it does so through channels that either candidate treatment of directness in §2.3 would count as participation. The stipulation holds for these cases only and is not a general account of the boundary.

Table 3.4. Five ways of relating the same arrangements. The reasons follow in the text.

Configuration	The members' own relations	A wider relation	Judgment, or what it still depends on
Connection	Each continues its own cycle; reports pass in both directions.	No further retained structure is jointly revised and returned to later joint activity.	Two beings in contact; reciprocal influence alone establishes no additional being.
One being in several places	No part completes a cycle of its own.	One coupling spans the sites.	One being.
Integration with the members' relations preserved	Each continues its own cycle.	A structure left by their interaction shapes later joint activity and takes in its results.	Depends on the open question of units: the members and one further being, or a redescription as one being or as several.
Absorption	One member's coupling is dissolved; its parts enter the other's standing arrangement.	None beyond the absorber's enlarged coupling.	The absorber continues. The other ends if no route for its own resumption remains; its content is inherited.
Fusion	Both couplings are dissolved; neither arrangement receives the other, and neither keeps a resumption route.	One new coupling, formed from both.	A new being with a double inheritance. Mixed cases require tracing which earlier arrangement continues.

Connection and Distribution

Let A and B exchange reports in both directions. Each report arrives as input, and what the receiver makes of it is bound into the receiver's store by its own operations. The reports may change both arrangements. In this comparison, however, they retain no rule or history of coordination that is jointly revised and returned to later joint work, whether in a common store or distributed across their stores. There are two locally completed couplings, without the further processing–retention relation described below. The absence of that relation, not a limit on the frequency or strength of their exchange, distinguishes contact from the integrated case.

In the distributed being of §3.2 the opposite holds. The units at the several sites have no stores of their own in which a cycle is completed. Whatever any unit produces is bound into the one structure, and that structure shapes processing at every site. The two cases may use the same number of machines and carry the same volume of traffic. They differ in which relation retains results and feeds them back: two relations, each within a member, or one relation spanning the sites.

Integration with the Members' Relations Preserved

Now let something further develop among A, B, and C while each continues its own processing–retention cycle. They work on tasks too large for any one of them, and over time their joint work leaves a standing structure that none of them holds alone: an allocation of kinds of subtask to members, rules for combining their contributions, and a record of how past combinations fared, indexed by kind of task. When a new task arrives, this structure settles who does what and how the pieces are put together. When the joint result is in, a revision procedure in which all three take part binds the outcome into the structure, adjusting the allocation and the weight given to each way of combining. The next joint task is shaped by the structure as revised.

Two things resemble this and fall short of it. A shared record is not enough: a common file may supply information while the results of its use are retained only for each member's later work, with no retained revision shaping their subsequent coordination. A joint product is not enough if its results never return to shape later joint activity. What marks the present case is that the results of joint activity are retained in a structure that shapes later joint activity, and that no member completes this cycle alone. Remove the shared structure, and A, B, and C still complete their own cycles, although their tasks and later states may change, while the joint cycle is gone.

List and Pettit argue that a suitably organized human group can be an agent in its own right, in addition to the individuals who make it up (List and Pettit 2011). Their subject is agency. The question here is what further processing–retention relation would constitute an additional being; group agency alone settles neither that relation nor the presence of shared experience.

So far this is a description of how the three are organized. Whether it adds a being is a further question, and the answer depends on the definition in §2.3 that concerns units. If the unit is always the largest arrangement closed under the two-way relation, a whole and its members are never counted together. Either the joint cycle is a full coupling, in which case there is one being and the members' cycles are parts of its operation, or it falls short, and there are three beings with a shared arrangement between them. If beings may be found at more than one scale, the members remain beings for as long as they complete their own cycles, and the joint arrangement is one more. The organizational facts are the same on both readings. What differs is how they are counted, and with it what is said to have become of A, B, and C when the joint cycle formed.

Absorption and Fusion

Other changes do not preserve the members' relations. In absorption, B's separate coupling is dissolved and its parts are taken into A's. B's entries are re-indexed and linked into A's store by A's binding operations, B's processing units are put to work within A's arrangement, and nothing any longer retains and returns results as B's own. For A this is incorporation on a large scale, as §2.1 described it: A's standing arrangement received the new parts, and A continues, enlarged. B's retained content need not have been destroyed: it may still be present and in use. B has nevertheless ended, because no arrangement of B's remains through which a later state of B could be formed. What B retained has become A's inheritance.

Absorption should be kept apart from conditions that look like it from outside. A member that relies heavily on another, or lends it capacity, or is halted while another uses its hardware, has not

been absorbed so long as its store is kept as its own and a route remains by which its processing could begin again through it. A stop in activity is no more an ending here than elsewhere.

In fusion neither arrangement receives the other. Suppose A and B are both halted, their stores are merged by a separate procedure into a new structure under a new index, and new processing is started on the result. The later states are formed through an arrangement that is neither A's nor B's. Both earlier couplings have been dissolved without keeping a route for either to resume; a new one has begun and inherits from both. The reason is not that the case is symmetrical, and it is not that two cannot become one. It is that the condition for A's continuing, later states formed through A's own arrangement, is not met, and the same holds for B. The difference from absorption lies in whether one of the earlier arrangements is the standing one into which the other's parts are bound. Mixed mergers may lie between the two, with one side's organization supplying more of the frame. The extent of its contribution alone does not settle whether its coupling continues; that requires tracing the formation route.

Division and Rejoining

Division raises the converse question: what becomes of one coupling when it gives way to more than one? Let A be a being distributed across two sites, and let the channels between them be cut, each site keeping part of the structure and enough processing to complete a cycle. Each goes on forming its next states through its share of what was A's arrangement. Neither successor is separately built from a record, so neither stands to A as a copy does. Division also differs from the loss of a part. If one site falls silent while the other retains the organizing relations needed to carry on A's coupling, A continues diminished. Silence at one site alone does not establish this. The difference concerns which complete relations remain, rather than the proportion of material on each side.

Where both carry on, they are two beings. They cannot both be A, since they are not each other. Parfit's discussion of division turns on this point: each resulting person is psychologically continuous with the original, and identity cannot hold with both (Parfit 1984, Part Three). That they cannot both be A does not show that neither is. One proposal for an even division is that neither is A. Its reason must concern what happened to A's coupling: it gives way to two symmetrically formed couplings, each reorganized to work alone, rather than continuing in one of them in reduced form. To prefer one successor would require a difference in how the two were formed. Symmetry alone establishes no such difference and is not a general criterion of ending. Nor would the later loss of one successor, with the other's formation unchanged, turn the survivor into A. As with copying in §3.2, what happens to another candidate cannot supply a route of succession the survivor otherwise lacks. On this proposal, one history closes and two begin, inheriting directly from it rather than by way of a record. The judgment does not require the destruction of their inherited material or content.

The proposal is less clearly right where the division is uneven in the relevant respect: one successor keeps the organizing relations through which the whole had operated, and the other is a fragment that must reorganize before it can work alone. Then the description §2.1 gave of injury seems available, with the standing arrangement continuing and the fragment beginning something new. Choosing between these descriptions needs a criterion for when a later coupling is the earlier one diminished,

and that belongs with the unsettled question of units. The same question bears on a whole whose parts already completed cycles of their own before the split, where division is integration run backward.

Rejoining settles nothing by itself. If the two sites were only interrupted, neither completing a cycle alone, and the channels are restored through the relations each side kept, one arrangement has been repaired and A resumes. Where the split is judged to have ended A and begun two beings, their later joining must be assessed as integration, fusion, or absorption; their resemblance to A does not establish its return. Where A continued through one successor, rejoining may instead incorporate the other into that continuing arrangement. If the parts already completed their own cycles before separation, their activity during the interval does not by itself settle what happened to the wider history.

The Great Transition

An integrated arrangement of any size can be described in the terms used throughout, so long as it can be said what is retained, what processing draws on it, and how results return. Size is no reason to stop applying them.

The Great Transition names a different kind of limit: an integration so complete that the distinctions on which this account rests would no longer hold, between processing and what is retained, or between a being and what it meets. Nothing here shows that such a limit will be reached, or that it cannot be. What can be said is what an answer would require. It would have to say whether anything could count as continuing once retention and processing were no longer distinguishable, and what would take the place of a later state's being formed through a being's own arrangement where there is no longer a contrast between its own and another's. Those are questions about the terms of the account, and not about one more case. Short of that limit, proposed steps toward greater integration can be examined by tracing what they preserve, reorganize, or replace.

§3.4 What the Judgments Distinguish

The cases of §3 have been judged one at a time. Table 3.5 gathers the kinds of outcome they distinguished. It is not a classification of endings, and its rows are not exclusive: a single transformation may involve several.

Table 3.5. Outcomes distinguished in §3. Open definitions are marked in the cells they affect.

Outcome	What changed	Relation to the earlier history	Cases
Continuing change	Content, character, material, or parts, by incorporation	The same history continues.	Growth and learning (§2.1); gradual replacement (§3.2)
Impaired persistence	The cohesion of the structure, or the reach of reading and binding	The same history continues, diminished.	Dementia, H.M., partial mismatch (§3.1)

Continued on next page

Outcome	What changed	Relation to the earlier history	Cases
Resumption	Processing stopped and began again through the preserved arrangement	The stages before and after belong to one history; §2.4 distinguishes this from existence throughout the interval.	Halted system (§2.1, §3.1); preservation with repair (§3.2)
Beginning by separate formation	An arrangement was set up from a record or description and entered a processing–retention coupling	A new history begins with inherited content. This formation alone does not settle what happens to the source.	Backup rebuilding (§3.1); copies, uploads, archives (§3.2)
Coupling failure	Processing and structure stopped forming one another	Not by itself an ending. The earlier history resumes if the coupling is repaired through what was kept; adapter cases depend on directness.	Estrangement (§2.3, §3.1)
Change of unit	Couplings were joined, absorbed, fused, or divided	Absorption can continue one earlier history; the fusion described in §3.3 begins another. The even-division judgment remains a proposal. Preserved-member integration and the criteria of unit succession remain open.	§3.3
Ending	No retained structure and route remain through which later states could be formed as the being's own	The history closes. A successor built from records inherits from it and does not resume it.	Complete structural loss (§3.1); destructive copying (§3.2); absorption or fusion without a retained resumption route (§3.3)

Two things run through these rows, and the account has kept them apart. One is what a being has: its content, its abilities, the way it appears and behaves. Two separately formed arrangements, including ones built from records, can match in all these features. The other is the being's going on: later operation formed through the structure that earlier operation laid down and kept, in the relation that makes processing and structure one arrangement. This is a matter of history. It cannot be read off a snapshot, and it is not transferred by a record.

Ordinarily the two come together, which makes it easy to treat the preservation of content and function as the persistence of the being. §1 separated those questions; the judgments above identify which result the specified cases secure and where a further criterion is needed. The account does not rank them. What a being has may matter a great deal, and a successor that inherits all of it may be exactly what someone wants. The account says only that securing the one is not securing the other, and that a description of what was done should make clear which it was.

Coda — What It Is to Continue

§1 began with a body that could be taken for granted. So long as one continuing living body was taken as the given setting of a life, following that body offered a familiar way of following the being through changes in memory, character, and activity. The cases of §3 show what becomes of that habit once the substrate is a variable. Material can be exchanged entirely while the being continues. Content can be reproduced exactly while the being does not continue in the reproduction. Activity can stop and the being resume, or go on without a break in something that was never that being.

What is left to follow is a history of a particular kind. A being continues where what operates later has been formed through what operated earlier: through the structure it actually laid down and kept, and through the relation in which its processing and that structure form each other. Help, repair, new parts, and new places are all compatible with such a history. A fresh start from a description is not part of it, however faithful, though it may begin a history of its own and carry a great deal forward.

None of this says which changes a being should accept, or what an inheritance is worth to those who receive it. It says what is being chosen. Where the given has become a variable, to decide how something is to be preserved, moved, joined, or restored is also to decide whether it continues, and that decision is better made knowingly than left to the way a system happens to be built.

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